FRESH CONCRETE PROPERTIES

Properties of fresh concrete:

- The potential strength and durability of concrete of a given mix proportion is very dependent on the degree of its compaction.
- It is vital, therefore, that the consistency of the mix be such that the concrete can be transported, placed, and finished sufficiently early enough to attain the expected strength and durability.

Significance:

- **The first 48 hours are very important for the performance of the concrete structure**.
- **It controls the long-term behavior, influence (ultimate strength), Ec (elastic modulus), creep, and durability.**

Elasticity and Strength Of Concrete

- The elastic properties of materials are a measure of their resistance to deformation under an applied load (but the elastic strain is recovered when the load is removed).
- Strength usually refers to the maximum stress that a given kind of sample can carry.
- Understanding these properties and how they are measured is essential for anyone wishing to use materials

Main Prop. OF Fresh Concrete

Consistency	Workability	Segregation	Bleeding
• Slump Test • Flow Test • Penetration Test	• Compacting Factor Test • VeBe Time Test	• --- • ---	• Bleeding Water Test

Consistency, is degree of wetness of concrete. Means how much wet the concrete is.

Workability means the ease with which concrete can be moved and placed in forms without segregation and bleeding .

Concrete Consistency

- Consistency or fluidity of concrete is an important component of workability and refers in a way to the wetness of the concrete.
- However, it must not be assumed that the wetter the mix the more workable it is. If a mix is too wet, segregation may occur with resulting honeycomb, excessive bleeding on the formed surfaces.

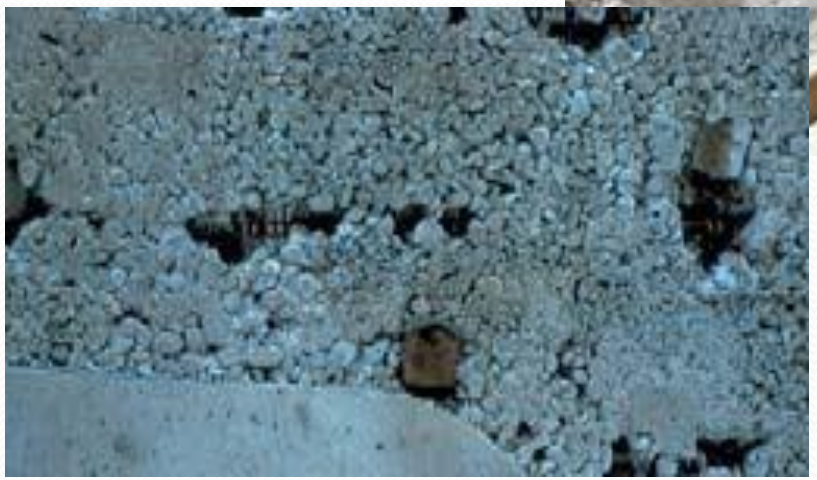




Concrete Consistency

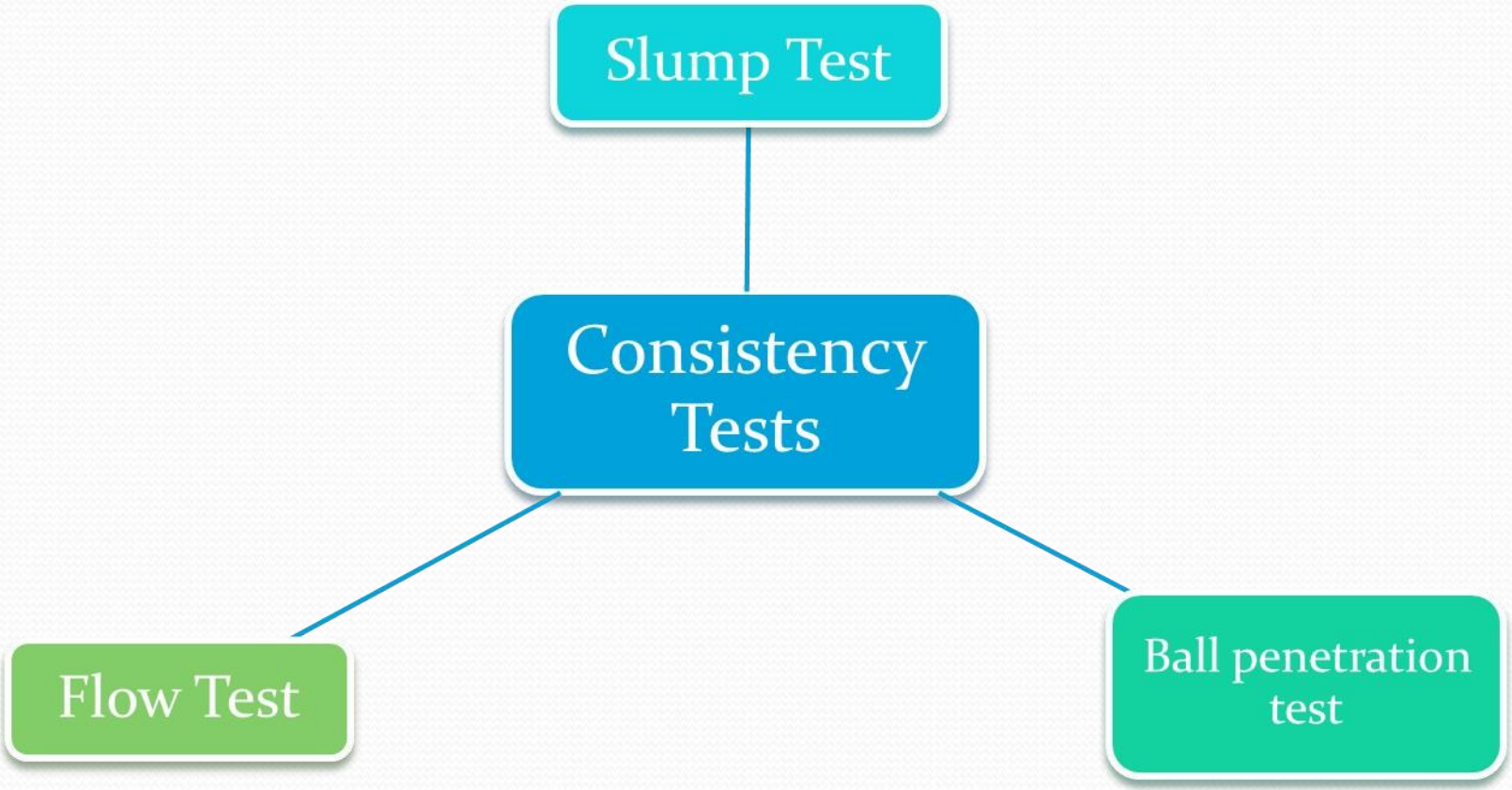
- On the other hand, if a mix is too dry it may be difficult to place and compact, and segregation may occur because of lack of cohesiveness and plasticity of the paste.







3 Ways to determine Consistency of Fresh Concrete



Slump Test

Definition

A slump test is a method used to determine the consistency of concrete. The consistency, or stiffness, indicates how much water has been used in the mix. The stiffness of the concrete mix should be matched to the requirements for the finished product quality

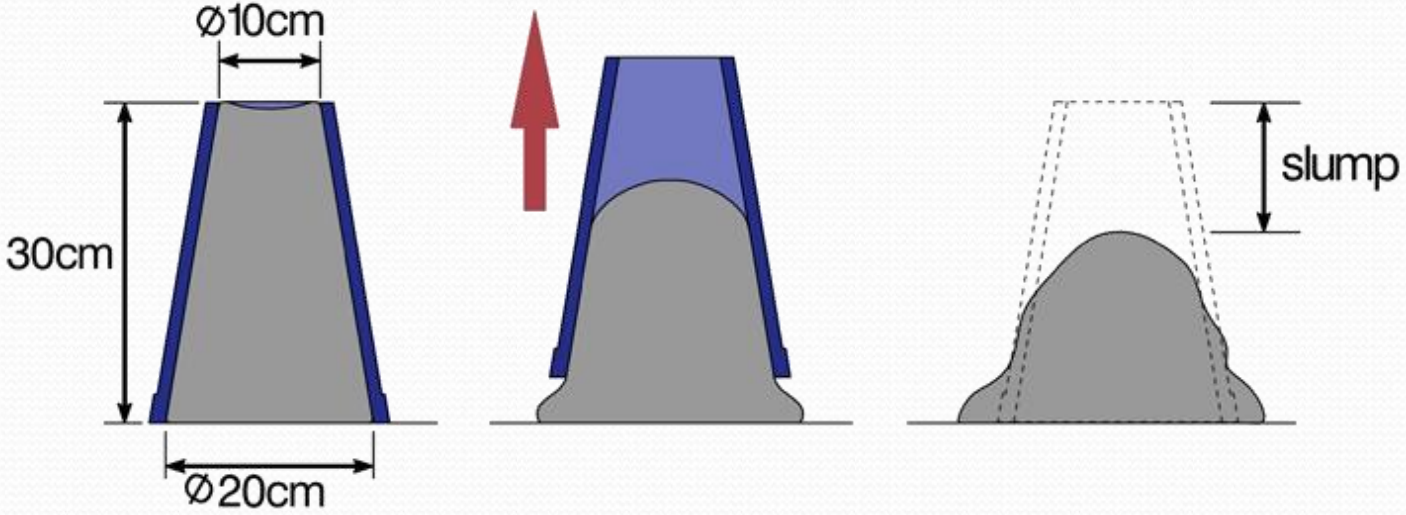
- Slump is a measurement of concrete's workability or fluidity.
- It's an indirect measurement of concrete consistency or stiffness.

Principle

The slump test result is a measure of the behavior of a compacted inverted cone of concrete under the action of gravity. It measures the consistency or the wetness of concrete.

Slump Test

- **Apparatus**
  - Slump cone : frustum of a cone, 300 mm (12 in) of height. The base is 200 mm (8in) in diameter and it has a smaller opening at the top of 100 mm
  - Scale for measurement,
  - Temping rod(steel) 15mm diameter, 60cm length.

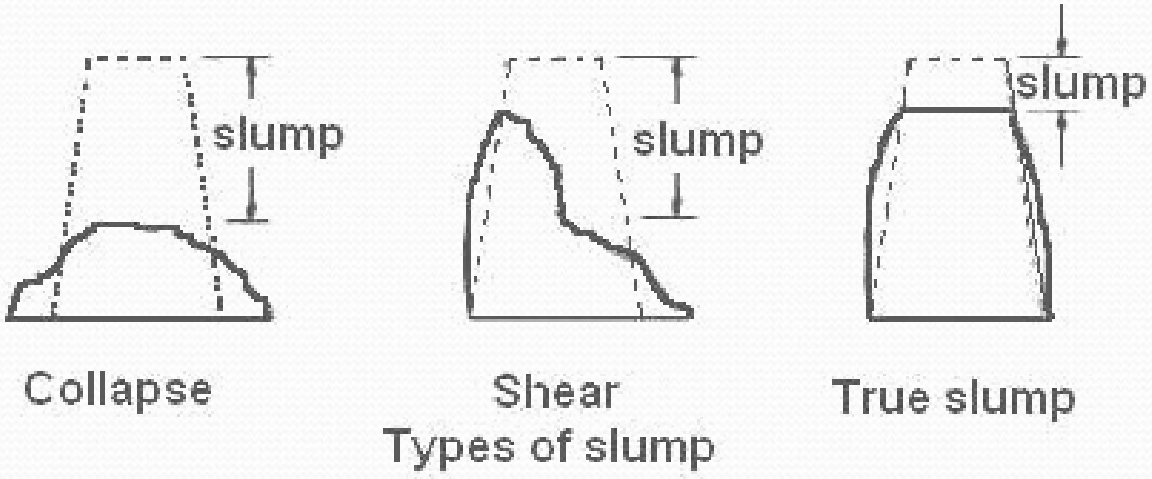


Slump Test

Types Of Slump

The slumped concrete takes various shapes, and according to the profile of slumped concrete, the slump is termed as;

- Collapse Slump
- Shear Slump
- True Slump



Slump Test

Types Of Slump

Collapse Slump

In a collapse slump the concrete collapses completely.

- A collapse slump will generally mean that the mix is too wet or that it is a high workability mix, for which slump test is not appropriate.

Shear Slump

In a shear slump the top portion of the concrete shears off and slips sideways. OR

If one-half of the cone slides down an inclined plane, the slump is said to be a shear slump.

- If a shear or collapse slump is achieved, a fresh sample should be taken and the test is repeated.
- If the shear slump persists, as may the case with harsh mixes, this is an indication of lack of cohesion of the mix.

Types Of Slump

True Slump

In a true slump the concrete simply subsides, keeping more or less to shape

- This is the only slump which is used in various tests.
- Mixes of stiff consistence have a Zero slump, so that in the rather dry range no variation can be detected between mixes of different workability.

However , in a lean mix with a tendency to harshness, a true slump can easily change to the shear slump type or even to collapse, and widely different values of slump can be obtained in different samples from the same mix; thus, the slump test is unreliable for lean mixes.

Degree of workability	Slump (mm)	Compacting Factor	Use for which concrete is suitable
Very low	0 - 25	0.78	Very dry mixes; used in road making. Roads vibrated by power operated machines
Low	25 - 50	0.85	Low workability mixes; used for foundations with light reinforcement. Roads vibrated by hand operated Machines
Medium	50 - 100	0.92	Medium workability mixes; manually compacted flat slabs using crushed aggregates. Normal reinforced concrete manually compacted and heavily reinforced sections with vibrations
High	100 - 175	0.95	High workability concrete; for sections with congested reinforcement. Not normally suitable for vibration

Slump Test

Slump (mm)	0 - 20	20 - 40	40 - 120	120 - 200	200 - 220
Consistency	Dry	Stiff	Plastic	Wet	Sloppy

>Table : Relation between Consistency and Slump values

Flow Test

The **flow table test** or **flow test** is a method to determine the consistence of fresh concrete.

- **Application :**

When fresh concrete is delivered to a site by a truck mixer it is sometimes necessary to check its consistence before pouring it into formwork.

- If the consistence is not correct, the concrete will not have the desired qualities once it has set, particularly the desired strength.

If the concrete is too pasty, it may result in cavities within the concrete which leads to corrosion of the rebar, eventually leading to the formation of cracks (as the rebar expands as it corrodes) which will accelerate the whole process, rather like insufficient concrete cover. Cavities will also lower the stress the concrete is able to support.

Flow Test

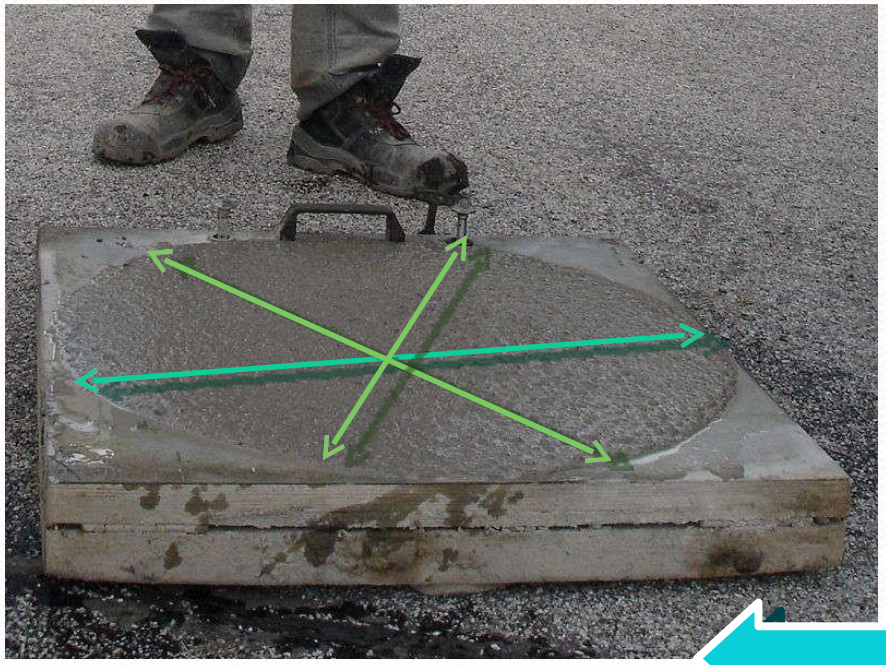
- **Equipment**
  - Flow table with a grip and a hinge, 70 cm x 70 cm.
  - Abrams cone, open at the top and at the bottom - 30 cm high, 17cm top diameter, 25 cm base diameter
  - Water bucket and broom for wetting the flow table.
  - Tamping rod, 60 cm height
  - Scale for measurement



Flow Test

- **Conducting**
  - The flow table is wetted.
  - The cone is placed on the flow table and filled with fresh concrete in two layers, each layer 25 times tamp with tamping rod.
  - The cone is lifted, allowing the concrete to flow.
  - The flow table is then lifted up several centimeters and then dropped, causing the concrete flow a little bit further.
  - After this the diameter of the concrete is measured in a 6 different direction and take the average.

Flow Test



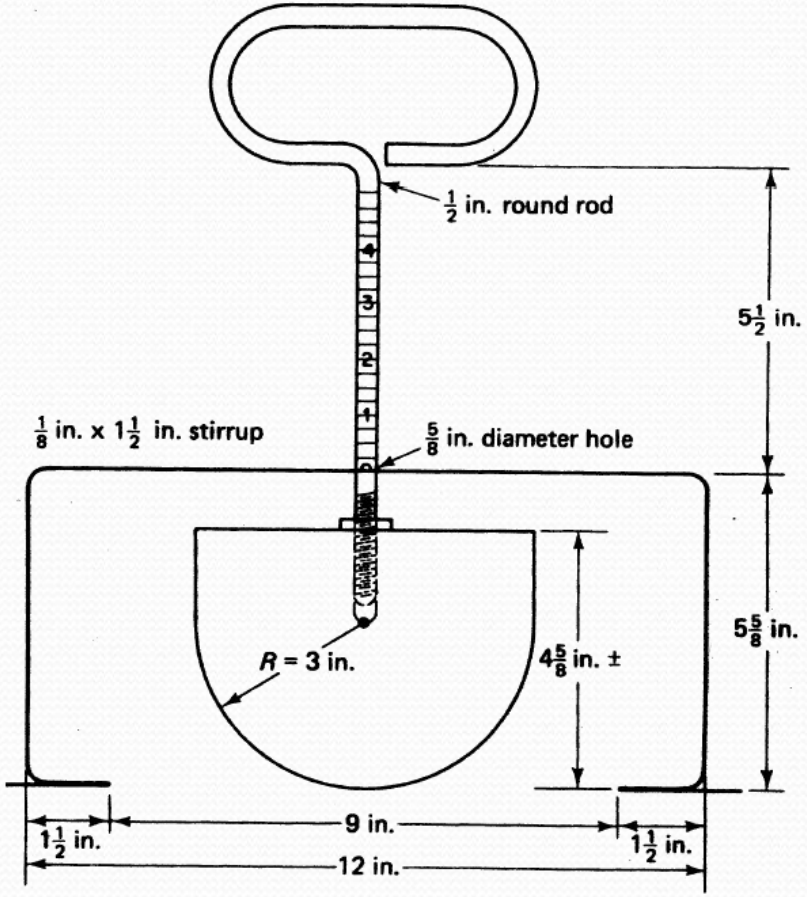


Flow Test

$$\textbf{\textit{Flow}}\ \% = \frac{Diameter\ of\ flow\ (cm) - 25}{25} \times 100$$

Percent of Flow	0 – 20 %	20 – 60 %	60 – 100 %	100 – 120 %	120 – 150 %
Consistency	Dry	Stiff	Plastic	Wet	Sloppy

Ball Penetration Test (Kelly Ball)





Ball Penetration Test (Kelly Ball)

Definition

Another method used in the field and laboratory to measure the consistency of concrete is the ball penetration test (ASTM C360) which is also known as the **Kelly ball test***.

Procedure

It is performed by measuring the penetration, in inches, of a 6-in. diameter steel cylinder with a hemi spherically shaped bottom , weighing 30 lbs.

Ball Penetration Test (Kelly Ball)

- **Advantages**
  - One of the advantages of the ball penetration test can be performed on the concrete in a hopper, buggy, wheelbarrow, or other suitable container.
  - Another advantage of this method is its simplicity and the rapidity with which the consistency of the concrete can be determined.
  - It is also not dependent on a procedure of filling and rodding a container like the slump test.

Concrete Workability

- **Definition:** Workability is often referred to as the ease with which a concrete can be transported, placed and consolidated without excessive bleeding or segregation
  - The property of fresh concrete which is indicated by the amount of useful internal work required to fully compact the concrete without bleeding or segregation in the finished product.
  - **Workability** is one of the physical parameters of concrete which affects the strength and durability as well as the cost of labor and appearance of the finished product
  - **Concrete is said to be workable when it is easily placed and compacted homogeneously i.e without bleeding or Segregation**. Unworkable concrete needs more work or effort to be compacted in place, also honeycombs &/or pockets may also be visible in finished concrete.

Concrete Workability

- **Factors affecting workability**
  - Water content in the concrete mix
  - Amount of cement &its Properties
  - Aggregate Grading (Size Distribution)
  - Nature of Aggregate Particles (Shape, Surface Texture, Porosity etc.)
  - Temperature of the concrete mix
  - Humidity of the environment
  - Mode of compaction
  - Method of placement of concrete
  - Method of transmission of concrete

Concrete Workability

- **How To improve the workability of concrete**
  - increase water/cement ratio
  - increase size of aggregate
  - use well-rounded and smooth aggregate instead of irregular shape
  - increase the mixing time
  - increase the mixing temperature
  - use non-porous and saturated aggregate
  - with addition of air-entraining mixtures

An on site simple test for determining workability is the SLUMP TEST.

Compacting Factor Test

Introduction

- These tests were developed in the UK by Glanville (1947) and it is measure the degree of compaction For the standard amount of work and thus offer a direct and reasonably reliable assessment of the workability Of concrete .
- The test require measurement of the weight of the partially and fully compacted concrete and the ratio the partially compacted weight to the fully compacted weight, which is always less than one, is known as compacted factor .
- For the normal range of concrete the compacting factor lies between 0.8 - 0.92

Compacting Factor Test

- **Apparatus**
  - Trowels
  - Hand Scoop (15.2 cm long)
  - Rod of steel or other suitable material (1.6 cm diameter, 61 cm long rounded at one end).
  - Balance.



Compacting Factor Test

$$\text{The Compacting Factor } (CF) = \frac{\text{weight of partially compacted concrete}}{\text{weight of fully compacted concrete}}$$

Workability	Slump (mm)	C.F	Uses
Very Low	0 - 25	0.78	Roads - Pavements
Low	25 - 50	0.85	Foundations Concrete
Medium	25 - 100	0.92	Reinforced Concrete
High	100 - 175	0.95	Reinforced Concrete (High Reinforcement)

VeBe Time Test

Definition

- It is based on measuring the time (Called VEBE time) needed to transfer the shape of a concrete mix from a frustum cone to a cylinder (these shapes are standardized by the apparatus of this test), by vibrating and compacting the mix.
- The more VEBE time needed the less workable the mix is. This method is very useful for stiff mixes.

Apparatus

- Cylindrical container with diameter = 240 mm, and height=200 mm
  Mold: the same mold used in the slump test.
- Disc : A transparent horizontal disc attached to a rod which slides vertically
- Vibrating Table : 380*260 mm, supported by four rubber shock absorbers
  Tamping Rod
- Stop watch

VeBe Time Test



1) The time required for the shape of concrete to change from slump cone shape to cylindrical shape in second is known as Vibe Degree.
2) This method is very suitable for very dry concrete whose slump value cannot be measure by slump test, but the vibration is too vigorous for concrete with slump greater than about 50m.

The test fails if VeBe Time is less than 5 seconds .. And the test must be created when no collapse or shears slump in concrete

Workability Description	Vee-bee Time (Seconds)
Extermely Dry	32 – 18
Very Stiff	18 – 10
Stiff	10 - 5
Stiff Plastic	5 – 3
Plastic	3 - 0

Concrete Segregation

Definition

- Segregation is when the coarse and fine aggregate, and cement paste, **become separated**. Segregation may happen when the concrete is mixed, transported, placed or compacted





Segregation makes the concrete

- WEAKER,
- LESS DURABLE,
- and will leave A POOR SURFACE FINISH

Concrete Segregation

Basic types of segregation

- *Coarse segregation* : Occurs when gradation is shifted to include too much coarse aggregate and not enough fine aggregate.
- Coarse segregation is characterized by low asphalt content, low density, high air voids, rough surface texture, and accelerated rutting and fatigue failure (Williams et. al., 1996b).
- Typically, coarse segregation is considered the most prevalent and damaging type of segregation; thus segregation research has typically focused on coarse segregation. The term "segregation" by itself is usually taken to mean "coarse segregation."

- *Fine segregation* : Occurs when gradation is shifted to include too much fine aggregate and not enough course aggregate. High asphalt content, low density, smooth surface texture, accelerated rutting, and better fatigue performance characterize fine segregation (Williams, Duncan and White, 1996).

Concrete Segregation

To Avoid Segregation

- Check the concrete is not 'too wet' or 'too dry'.
- Make sure the concrete is properly mixed. It is important that the concrete is mixed at the correct speed in a transit mixer for at least two minutes immediately prior to discharge.
- The concrete should be placed as soon as possible.
- When transporting the mix, load carefully.
- Always pour new concrete into the face of concrete already in place.
- When compacting with a poker vibrator be sure to use it carefully

Concrete Segregation

To Avoid Segregation

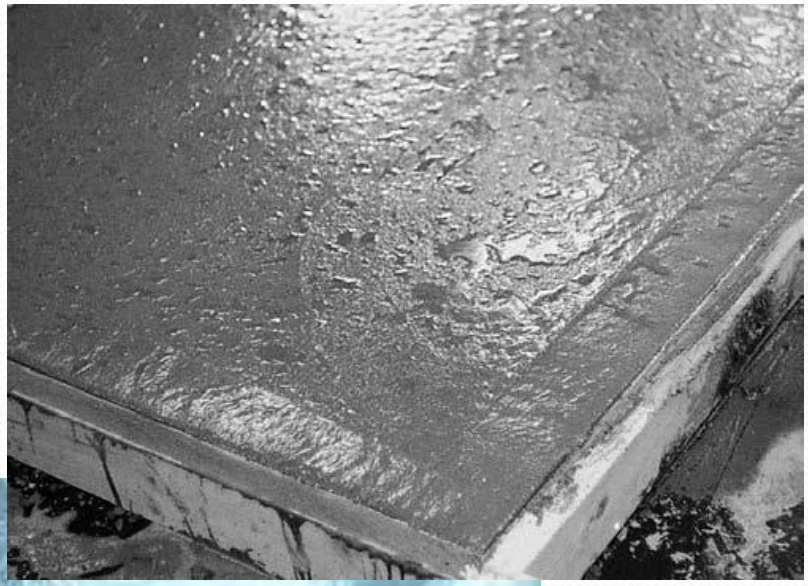
- If placing concrete straight from a truck, pour vertically and never let the concrete fall more than one-and-a-half meters.





Concrete Bleeding







Concrete Bleeding

- This refers to the appearance of water along with cement particles on the surface of the freshly laid concrete.
- This happens when there is excessive quantity of water in the mix or due to excessive compaction.

- Bleeding causes the formation of pores and renders the concrete weak. Bleeding can be avoided by suitably controlling the quantity of water in the concrete and using finer grading of aggregates.
- A thorough knowledge of why concrete bleeds and how mix proportions affect it, is required to preventing the harmful effects of bleeding.
- Adoption of right finishing methods also helps to ensure that the bleeding problems won't ruin a slab surface.

How to control bleeding

- Don't add too much water to the concrete. Most of the water added to make placing easier bleeds out of the concrete.
- Place concrete at the lowest possible slump. If you need a higher slump to speed placement, consider using a super plasticizer.
- Add additional concrete fines to reduce bleeding. The fines may come from a number of sources:
- Use a more finely ground cement. Concretes made with high early strength (Type III) cement bleed less because the cement is ground finer than normal (Type I) cement.
- Use more cement. At the same water content, rich mixes bleed less than lean mixes.
- Use fly ash or other pozzolanas in the concrete.
- For air- entrained concrete, use the maximum allowable amount of entrained air. Consider using an air- entraining agent whenever excessive bleeding is a problem. Entrained air bubbles act as additional fines. Air entrainment also lowers the amount of water needed to reach a desired slump.